Number Theory

Lecture 1

20 January, 2025

1 Peano's Axioms:

1. There exists a positive integer 1.
2. If x is a positive integer, then so its successor x'.
3. There is no positive integer whose successor is 1.
4. If $x' = y'$, then $x = y$.
5. If a set S of positive integers contains 1 and successors of each of its members, then S contains all positive integers.

1.1 Definition

$$\begin{aligned} x + 1 &= x' \\ x + y' &= (x+y)' \end{aligned}$$

1.2 Definition

$$\begin{aligned} x \cdot 1 &= x \\ x \cdot y' &= x \cdot y + x \end{aligned}$$

Exercise. Prove that 3+2=5.

Solution.

$$\begin{aligned} 3 + 2 &= 3 + 1' \\ &= (3+1)' \\ &= (2'+1)' \\ &= ((2+1)')' \\ &= ((1'+1)')' \\ &= ((1+1')')' \\ &= 1'''' = 5 \end{aligned}$$

2 Well Ordering Principle:

Every non-empty set of positive integers has a least element.

Proof. Let S be a non-empty subset of positive integers with no least element.

Define a set $T = \{n \text{ is a positive integer} \mid n < x \text{ for all } x \in S\}$
Since S has no least element, 1 cannot be an element of S.
Therefore, 1 is less than every element in S, so $1 \in T$.
Assume that, $k \in T$. This means $k < x$ for all $x \in S$.
If $k+1 \in S$, it would be the least element of S (contradiction).
Therefore, $k+1$ must also be less than every element in S, implying $k+1 \in T$.

So, using the induction axiom, we can conclude that T contains all the positive integers.
Since, $T \cap S = \phi$ so it implies that S must be empty also.
This contradicts our assumption that S is non-empty. Hence the result follows.

3 Generalized Well Ordering Principle:

Suppose, a is any integer and S is the set of positive integers greater than or equal to a. Then any non-empty subset of S has a least element.

Proof. left as an exercise.

4 Division Algorithm:

4.1 If a, b are integers with $b > 0$, then there exist unique integers q and r such that $a = bq + r\ ,\ 0 \leq r < b$.

Proof.
Take, $S = \{a - bx \mid x \text{ is an integer and } a - bx \geq 0\}$

We take $x = -|a|$ so that $a - bx = a + |a|b \geq 0$, hence $S \neq \phi$.

Then by Well Ordering Principle, S has a least element r.(say)
i.e. $r = a - bq$ for some integer q
i.e. $a = bq + r\ ,\ r \geq 0$

If $r \geq b$ then $r - b \geq 0$ and $r - b < r$
and $r - b = (a - bq) - b = a - (q+1)b = a - bx \in S$
— a contradiction to the fact that r is least in S.
Hence, $r < b$.

Suppose, $a = bq_1 + r_1 = bq_2 + r_2$ with $0 \leq r_1 < b$ and $0 \leq r_2 < b$.
Then, $b(q_1 - q_2) = r_2 - r_1$
so, $|b||q_1 - q_2| = |r_2 - r_1|$
As $|r_2 - r_1| < b$, hence $b|q_1 - q_2| < b$
So, $0 \leq |q_1 - q_2| \leq 1$
Thus, $|q_1 - q_2| = 0$.
i.e. $q_1 = q_2$
Thus, $r_1 = r_2$.

4.2 If a, b are integers with $b \neq 0$ then there exist unique integers q and r such that $a = bq + r$, $0 \leq r < |b|$.

Proof. If $b > 0$, then the result follows by previous result.
If $b < 0$, then $-b > 0$, so by previous result there exist unique integers q' and r' such that $a = (-b)q' + r'$, $0 \leq r' < -b$
i.e. $a = b(-q') + r'$, $0 \leq r' < |b|$
i.e. $a = bq + r$, $0 \leq r < |b|$, where $q = -q', r = r'$.

4.3 If a, b are integers with $b \neq 0$ then there exist unique integers q and r such that $a = bq + r$, $-\frac{1}{2}|b| < r \leq \frac{1}{2}|b|$.

Proof. left as an exercise.

Note

$\lfloor x \rfloor$ = greatest integer not exceeding x
So, $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$
i.e. $0 \leq x - \lfloor x \rfloor < 1$
i.e. $0 \leq F_r(x) < 1$, where $F_r(x) = x - \lfloor x \rfloor$ is the fractional part of x

Now, $x = \lfloor x \rfloor + F_r(x)$
By, division algorithm version 4.2 , $\exists\, q, r$ such that $a = bq + r$, $0 \leq r \leq |b|$.
$\Rightarrow \dfrac{a}{|\,b\,|} = \dfrac{b}{|\,b\,|}q + \dfrac{r}{|\,b\,|}$, with $0 \leq \dfrac{r}{|\,b\,|} < 1$
$\Rightarrow \dfrac{a}{|\,b\,|} = (\text{sgn } b)\; q + \dfrac{r}{|\,b\,|}$, where $\text{sgn } b = \begin{cases} 1 & \text{if } b > 0 \\ -1 & \text{if } b < 0 \end{cases}$, so $b\ (\text{sgn } b) = |b|$ and $(\text{sgn } b)^2 = 1$
Hence, $(\text{sgn } b)\; q = \left\lfloor \dfrac{a}{|\,b\,|} \right\rfloor$
$\Rightarrow q = \left\lfloor \dfrac{a}{|\,b\,|} \right\rfloor (\text{sgn } b)$

Thus,

$$\begin{aligned} r &= a - bq \\ &= a - b(\text{sgn } b)\left\lfloor \frac{a}{|\,b\,|} \right\rfloor \\ &= a - |b|\left\lfloor \frac{a}{|\,b\,|} \right\rfloor \end{aligned}$$

Note

By division algorithm version 4.1, for any positive integer a, $a = 4q + r$, $r = 0$ or 1 or 2 or 3
so, $a = 4q$ or $a = 4q + 1$ or $a = 4q + 2$ or $a = 4q + 3$
i.e. $a^2 = 16q^2$ or $a^2 = 16q^2 + 8q + 1$ or $a^2 = 16q^2 + 16q + 4$ or $a^2 = 16q^2 + 24q + 9$
i.e. $a^2 = 4q'$ or $a^2 = 4q' + 1$ or $a^2 = 4q'$ or $a^2 = 4q' + 1$
(q' may be different in each case)
Thus, **any square number can be written in the form of 4q or 4q + 1, where q is an integer.**